

ADVANCED MACHINE LEARNING & FULL-STACK MLOPS

Comprehensive Professional Curriculum

A holistic roadmap from data engineering and mathematical foundations to production-grade deployment.

MODULE 1: INTRODUCTION TO MACHINE LEARNING & PROJECT FRAMING

- **Foundations of AI:** Distinguishing Artificial Intelligence, Machine Learning, Deep Learning, and Generative AI paradigms.
- **ML Architectures & Learning Paradigms:** Batch vs. Online learning; Instance-based vs. Model-based architectures; introduction to high-dimensional tensors.
- **The Production Lifecycle (MLDLC):** End-to-end Machine Learning Development Life Cycle; defining roles across Data Engineering, Data Science, and DevOps/MLOps.
- **Problem Formulation & Framing:** Transforming business objectives into concrete ML tasks, performance metrics definition, and data viability assessment.

MODULE 2: DATA GATHERING & EXPLORATORY DATA ANALYSIS (EDA)

- **Data Ingestion Pipelines:** Programmatic extraction using structured formats (CSV, JSON, Parquet) and relational/non-relational databases.
- **Advanced Web Scraping & APIs:** Building custom scrapers for unstructured web data; authenticating and querying complex REST APIs.
- **Exploratory Data Analysis (EDA):** Structural validation, profiling, statistical distributions, and deep-dive Univariate, Bivariate, and Multivariate analysis using visualization frameworks.

MODULE 3: FEATURE ENGINEERING & PREPROCESSING PIPELINES

- **Missing & Categorical Data Treatment:** Complete Case Analysis, advanced statistical imputation, KNN Imputation, and Multivariate Imputation by Chained Equations (MICE).
- **Categorical Encoding:** Nominal vs. Ordinal mapping, One-Hot encoding optimization, Target encoding, and handling rare levels.
- **Mathematical Transformations & Scaling:** Box-Cox / Yeo-Johnson power transforms, log/reciprocal mapping; Standardization vs. MinMax normalization.
- **Outlier Management & Binning:** Interquartile Range (IQR), Z-score trimming, percentile capping, and feature discretization.
- **Production-Grade Workflows:** Packaging complex transformations into reproducible, deterministic Scikit-Learn Pipelines and custom ColumnTransformers.

MODULE 4: DIMENSIONALITY REDUCTION & CORE MATHEMATICAL FOUNDATIONS

- **The Curse of Dimensionality:** Geometric anomalies in high-dimensional spaces and computational bottlenecks.
- **Principal Component Analysis (PCA):** Geometric intuition, variance preservation, mathematical derivation via the Covariance Matrix, and Eigenvalue/Eigenvector decomposition from scratch.
- **Linear Algebra Fundamentals:** Vector spaces, matrix transformations, and matrix factorizations critical for optimization algorithms.

MODULE 5: SUPERVISED LEARNING – REGRESSION ALGORITHMS & OPTIMIZATION

- **Linear Regression Mechanics:** Simple and Multiple Linear Regression frameworks; full mathematical derivation of the Ordinary Least Squares (OLS) closed-form solution.
- **Mathematical Optimization via Gradient Descent:** Implementing Batch, Stochastic (SGD), and Mini-Batch Gradient Descent from first principles.
- **Regularization Strategies:**
 - Ridge Regression (L_2 Regularization) closed-form and iterative solutions.
 - Lasso Regression (L_1 Regularization) formulation, coordinate descent, and mathematical mechanics of feature sparsity.
 - ElasticNet optimization.
- **Evaluation Metrology:** Comprehensive analysis of MAE, MSE, RMSE, R^2 , and Adjusted R^2 .

MODULE 6: SUPERVISED LEARNING – CLASSIFICATION & NON-LINEAR MODELS

- **Logistic Regression & Perceptrons:** The Perceptron trick, activation functions, cross-entropy/log-loss optimization math; Multi-class classification (One-vs-Rest and Multinomial).
- **Instance-Based Learning:** K-Nearest Neighbors (KNN) architecture, distance metrics (Euclidean, Manhattan, Minkowski), and hyperparameter tuning.
- **Tree-Based & Ensemble Architectures:**
 - Decision Trees: Entropy, Information Gain, and Gini Impurity calculations.
 - Bagging & Random Forests: Bootstrapping mechanics, out-of-bag error evaluation, and variance reduction.
 - Boosting Paradigms: Concept of weak learners; detailed exploration of Gradient Boosting, AdaBoost, and modern frameworks (XGBoost, LightGBM).

MODULE 7: BACKEND ENGINEERING & API DEVELOPMENT

- **Application Frameworks:** Structuring enterprise backends utilizing FastAPI and Flask.
- **Model Serialization:** Safely exporting trained models, pipelines, and state weights using Joblib, Pickle, or ONNX formats.
- **API Architecture:** Designing RESTful endpoints to process incoming JSON payloads, run predictive inference through pipelines, and structure JSON responses.
- **Validation & Security:** Data serialization, input validation using Pydantic schemas, and error handling mechanisms.

MODULE 8: FRONTEND ENGINEERING FOR AI APPLICATIONS

- **Rapid Prototyping:** Building functional data applications quickly using Streamlit for internal testing or client demos.
- **Production Web Interfaces:** Developing fully responsive frontends using HTML5, CSS3 (modern layouts), and vanilla JavaScript or modern frontend frameworks.
- **Asynchronous State Management:** Executing async HTTP requests (Fetch API / Axios) to stream or send input variables to backend ML endpoints and dynamically render inference outputs.

MODULE 9: MLOPS, CONTAINERIZATION & PRODUCTION DEPLOYMENT

- **Containerization with Docker:** Writing optimal Dockerfiles, containerizing backend-frontend environments, and managing local multi-container services with Docker Compose.
- **Cloud Infrastructure & Deployment:** Architecting and executing deployments across major cloud platforms (AWS EC2/ECS, GCP Compute Engine, or Azure).
- **CI/CD Pipelines:** Implementing continuous integration and continuous deployment pipelines (GitHub Actions/GitLab CI) to automate model linting, testing, and delivery.
- **Production Monitoring & Reliability:** Setting up structured logging, performance health-checks, tracking data drift, and designing graceful rollback strategies.